Yixuan Wang

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EDUCATION

University of Florida — Ph.D., Mechanical & Aerospace Engineering 05/2024 – Present

- Nonlinear Controls and Robotics (NCR) Lab; advisor: Prof. Warren E. Dixon
- Research focus: nonlinear control, Lyapunov stability, generative modeling, reinforcement learning

Johns Hopkins University — M.S., Computer Science 01/2023 – 05/2024

- Concentration: Machine Learning & Natural Language Processing

University of Florida — M.S., Mechanical Engineering 08/2020 – 06/2022

- Major: Dynamics & System Control; Minor: ECE. GPA 3.75/4.0. Herbert Wertheim College of Engineering Achievement Award Scholarship.

Tongji University — B.S., Mechanical Design, Manufacture and Automation 09/2015 – 06/2020

PUBLICATIONS

1. **Effective Model Pruning: Measuring the Redundancy of Model Components.**
   Yixuan Wang, Dan Guralnik, Saiedeh Akbari, Warren E. Dixon.
   ICML 2026 (Spotlight). arXiv:2509.25606
2. **Goal Inference with Rao-Blackwellized Particle Filters.**
   Yixuan Wang, Dan P. Guralnik, Warren E. Dixon.
   IFAC World Congress 2026. arXiv:2512.09269
3. **Quantifying Trade-Offs Between Stability and Goal-Obfuscation.**
   Yixuan Wang, Dan Guralnik, Warren E. Dixon.
   arXiv preprint, 2026. arXiv:2605.06630
4. **Energy Generative Modeling: A Lyapunov-based Energy Matching Perspective.**
   Yixuan Wang, Wenqian Xue, Warren E. Dixon.
   arXiv preprint, 2026. arXiv:2605.05530
5. **Riemannian Lyapunov Optimizer: A Unified Framework for Optimization.**
   Yixuan Wang, Omkar Sudhir Patil, Warren E. Dixon.
   arXiv preprint, 2026. arXiv:2601.22284
6. **Lyapunov-Based Kolmogorov-Arnold Network (KAN) Adaptive Control.**
   Xuehui Shen, Wenqian Xue, Yixuan Wang, Warren E. Dixon.
   arXiv preprint, 2025. arXiv:2512.21437
7. **Off-Dynamics Reinforcement Learning via Domain Adaptation and Reward Augmented Imitation.**
   Yihong Guo, Yixuan Wang, Yuanyuan Shi, Pan Xu, Anqi Liu.
   NeurIPS 2024. arXiv:2411.09891

SKILLS

Programming: Python, MATLAB / Simulink, C/C++

Research: Nonlinear control & Lyapunov analysis, generative modeling, reinforcement learning, Bayesian estimation, model compression, robotics

Languages: English (proficient), Mandarin (native), German (B2)